Solution Architecture

Date 29 september 2026
Team ID 6ab2ac2a1266c775575a9e56
Project Name Farm Management System
Maximum Marks 5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

[e.g., Web AWeb Application

Farmer Dashboard

Admin Dashboard

pp (React), Mobile App (Flutter), Admin Dashboard]

HTML, CSS, JavaScript / React

↓ ↑

API Gateway / Load Balancer

REST API

Node.js + Express.js

↓ ↑

Auth Service *User Registration*

User Login

Authentication

Role Management

Core Logic Service

Farm Management

Crop Management

Field Management

Irrigation Management

Fertilizer Management

Pest/Disease Management

Farm Reports

↓ ↑

Data / Storage Layer

External API Integrations Weather API

[e.g., PostgreSQL Database, MongoDB, AWS S3]





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	*Provides user interface for farmers and administrators to register, login, manage farms, crops and view farm information.*	*HTML, CSS, JavaScript / React*
API Gateway	*Receives client requests and routes them to the required backend services.*	*Node.js, Express.js, REST API*

Microservices / Core Services	*Handles authentication, farm management, crop management, irrigation, fertilizer and pest/disease management.*	*Node.js, Express.js*

Database Stores user, farm, field, crop and management-related information securely. MongoDB / MySQL